

Research paper

Design, modeling, and manufacturing of delta wings with wing fences

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ARTICLE INFO

Keywords:

Delta wing
Wing fences
UAV modeling
UAV manufacturing
Aerodynamic stability

ABSTRACT

This work presents the aerodynamic design, structural integration, and experimental validation of a low-cost delta-wing UAV optimized for high-altitude, low-density environments such as Bogotá, Colombia. A multi-stage workflow was implemented, comprising computational fluid dynamics (CFD) analysis, finite element verification, and telemetry-based flight testing using IMU sensors. Fifteen airfoils were numerically screened under both gliding and powered conditions, identifying the EPPLER 396 profile as the most efficient, achieving a maximum lift-to-drag ratio of $C_L/C_D = 71.2$ and a stabilizing moment slope $dC_M/d\alpha = -0.045$. The addition of passive flow-control devices (winglets and wing fences) reduced induced drag by approximately 8% and increased roll damping by 32%, as confirmed through five repeated 300 s flight tests. The measured longitudinal damping ratio ($\zeta_L = 0.36 \pm 0.04$) and roll dispersion ($\sigma_\phi = 2.9^\circ$) demonstrated stable flight behavior across all tests. The results are consistent with recent findings, validating the aerodynamic benefits of spanwise fences for low-Reynolds delta configurations. Overall, the integrated CFD-experimental approach provides a reliable and accessible framework for aerodynamic validation of small-scale UAVs operating in high-altitude environments.

1. Introduction

Small radio frequency-controlled aircraft have their origins in military operations, where the use of manned vehicles posed significant risks to personnel. These unmanned aerial vehicles (UAVs), often designed as flying wings, emerged as cost-effective and operationally flexible alternatives that allowed missions to be conducted without the inherent dangers of human flight [1,2]. Over time, their applications have expanded to civilian domains such as logistics, precision agriculture, meteorological data collection, aerial mapping, and environmental monitoring [3–6]. This transition illustrates the adaptability of flying-wing configurations to diverse missions and flight envelopes.

Within this context, airfoil selection plays a critical role in defining aerodynamic efficiency and stability. Airfoils are fundamental to all aerodynamic platforms—from commercial aircraft to gliders and micro UAVs—and their selection depends on mission-specific parameters such as cruising velocity, Reynolds number, payload, and desired flight autonomy [6,7]. For flying wings, in particular, the absence of a conventional tailplane increases sensitivity to aerodynamic loads; therefore, airfoils must exhibit high lift coefficients at moderate angles of attack

to ensure pitch and roll stability in turbulent or low-density environments [8,9].

The aerodynamic performance of an airfoil is primarily characterized by the lift (C_L) and drag (C_D) coefficients. Their ratio, the lift-to-drag efficiency (C_L/C_D), serves as a primary metric for aerodynamic optimization [7]. Achieving a high C_L/C_D requires maximizing lift without inducing excessive viscous or pressure drag, which directly influences endurance and energy consumption [9,10]. In modern aerodynamic design, computational fluid dynamics (CFD) has become a fundamental tool for predicting these parameters and guiding optimization.

Recent advances in computational optimization and artificial intelligence have significantly improved airfoil design. Vinuesa et al. [11] applied deep reinforcement learning for active flow control in low-aspect-ratio wings, while Robb and Paul [12] demonstrated spanload optimization for improved flying-wing stability. Similarly, Lian et al. [13] reviewed the progress of evolutionary algorithms in aerodynamic design, and subsequent contributions have explored multi-objective strategies and learning-based surrogates under low-Reynolds regimes [14–16]. These studies highlight a paradigm shift toward hybrid approaches that combine CFD, surrogate modeling, and data-driven control.

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<https://doi.org/10.1016/j.rineng.2025.108857>

Received 29 July 2025; Received in revised form 26 October 2025; Accepted 22 December 2025

Available online 30 December 2025

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Fig. 1. Illustration of winglet-induced vortex control. Adapted from [16].

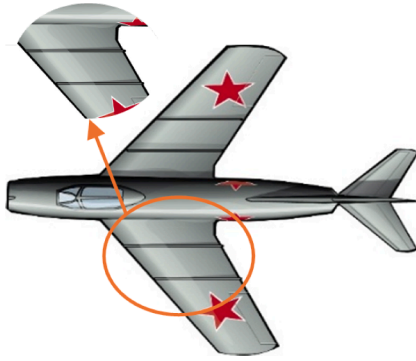


Fig. 2. Schematic of wing-fence placement on a swept/delta planform. Adapted from [16].

Flying wings, however, present unique aerodynamic challenges due to the absence of horizontal stabilizers. Their highly swept geometries and center-of-pressure variability can result in pitch oscillations and lateral instability [10,15]. To counteract these effects, passive control surfaces such as winglets and wing fences are widely used. Winglets-vertical extensions at the wingtips-reduce induced drag by mitigating wingtip vortices (Fig. 1) [16].

Wing fences (or flow correctors), depicted in Fig. 2, are flat vertical surfaces placed along the span to limit spanwise flow migration, delay boundary-layer separation, and enhance both longitudinal and lateral stability. Recent investigations-ranging from experimental/parametric analyses to large-eddy simulations-have reported improvements in post-stall behavior, drag reduction, and effective damping in delta-wing UAVs equipped with fences [16,17].

Delta-wing configurations, frequently adopted in UAVs and micro air vehicles, rely on specialized airfoils capable of maintaining attached flow at high angles of attack. Profiles such as WORTMANN FX 63-137 and A18 provide favorable pressure gradients that delay flow separation, whereas others like Clark Y, DF101, and MH45 use camber modulation to balance lift and static stability (Fig. 3). Selecting an optimal profile thus requires balancing aerodynamic efficiency, manufacturability, and stability within the expected Reynolds range [6,7,18].

From a structural perspective, the selection of lightweight yet resilient materials remains crucial. Although carbon composites are increasingly common, polystyrene and balsa wood continue to dominate in experimental and educational UAVs due to their low cost, ease of shaping, and favorable strength-to-weight ratio. Nevertheless, such materials demand reinforcement-typically through spars or composite mesh-to resist bending and torsional stresses during dynamic maneuvers [19–22]. Integrating aerodynamic optimization with structural design enables low-cost yet robust solutions suitable for experimental validation.

Despite extensive advances in CFD and optimization techniques, there remains a scarcity of studies that experimentally validate aerodynamic stability improvements in flying-wing UAVs operating under low-density, high-altitude conditions. Furthermore, few works integrate numerical optimization, passive control devices (such as wing fences), and real-world telemetry-based validation. Addressing these gaps is fun-

damental to understanding the aerodynamic and structural coupling effects governing stability in small-scale UAVs.

This study presents the design, numerical modeling, manufacturing, and flight validation of a delta-wing UAV optimized for the atmospheric conditions of Bogotá, Colombia-characterized by high altitude (≈ 2550 m) and reduced air density. The workflow comprises (i) aerodynamic profile selection and CFD-based evaluation under local conditions, (ii) iterative fuselage and wing-fence integration to enhance static and dynamic stability, and (iii) experimental validation using IMU-based telemetry with statistical analysis of flight performance. This multidisciplinary approach aims to provide an accessible, data-driven methodology for the aerodynamic design of low-cost UAVs operating in low-Reynolds environments.

2. Methods

2.1. Overview and workflow

The methodological workflow followed in this research is summarized in Fig. 4. It comprises four main stages: (i) airfoil screening and aerodynamic analysis, (ii) parametric design of the fuselage and integration of electronic components, (iii) CFD-based evaluation of passive flow devices (winglets and fences), and (iv) manufacturing and experimental validation through repeated flight tests. This structure ensures both computational consistency and experimental reproducibility.

2.2. Airfoil screening and selection

Twenty representative airfoils from the Wortmann, Eppler, Clark, and Boeing families were initially evaluated using XFLR5 to ensure a wide coverage of thickness-to-chord ratios, camber lines, and Reynolds-number adaptability within the range 1.4×10^5 – 7.4×10^5 . Although the Wortmann family was expected to perform well under low Reynolds numbers, the broader set was included to **avoid bias and to confirm the selection through comparative aerodynamic evidence** rather than prior assumptions. The screening criteria were the lift-to-drag ratio (C_L/C_D) and pitching-moment stability (C_M). To avoid prior-family bias and support selection by comparative evidence rather than expectation, the initial set included Wortmann, Eppler, Clark and Boeing geometries spanning thickness/camber variations across $Re \in [1.4, 7.4] \times 10^5$. Selection was based on $E = C_L/C_D$ and on a negative and sufficiently steep $dC_m/d\alpha$.

2.3. Design and integration

The selected airfoil (EPPLER 396) was integrated into a flying-wing configuration designed parametrically in SolidWorks. The fuselage geometry evolved iteratively to minimize drag and optimize internal volume, maintaining the center of gravity within 20%–25% of the mean aerodynamic chord. A 3DR Pixhawk IMU was included in the design layout to measure pitch, roll, yaw, altitude, and airspeed. These measurements were recorded via the onboard telemetry system and later used to validate the predicted aerodynamic stability.

2.4. CFD simulation and flow control devices

CFD analyses were performed in ANSYS Fluent 2024 R2 under steady-state and transient regimes. The realizable $k-\epsilon$ model was adopted to capture turbulent dissipation at low Reynolds numbers. Mesh-independence was verified using three refinement levels (0.6–1.5 million elements), with convergence reached when residuals dropped below 10^{-5} and aerodynamic coefficients stabilized within 1%. Winglets and fences were introduced parametrically following NACA's classical induced-drag minimization guidelines. Their geometry was adjusted to deflect the tip vortex core 10–12% spanwise, targeting a reduction in induced drag and improvement in pitching stability. Telemetry from the

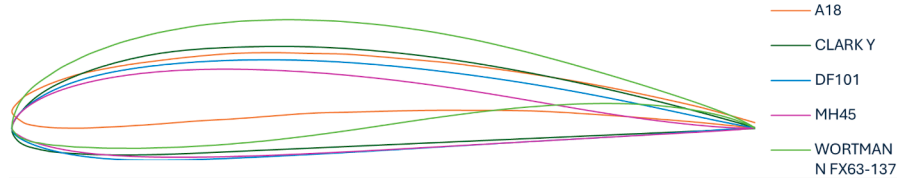


Fig. 3. Representative airfoils considered for low-Reynolds-number UAV applications. Source: [18].

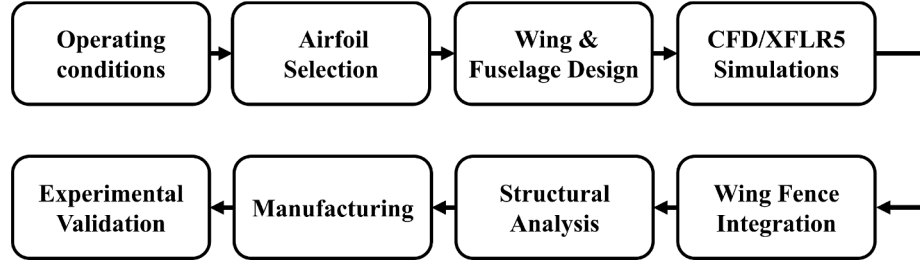


Fig. 4. Workflow of the study: from airfoil selection and design to CFD evaluation and experimental validation.

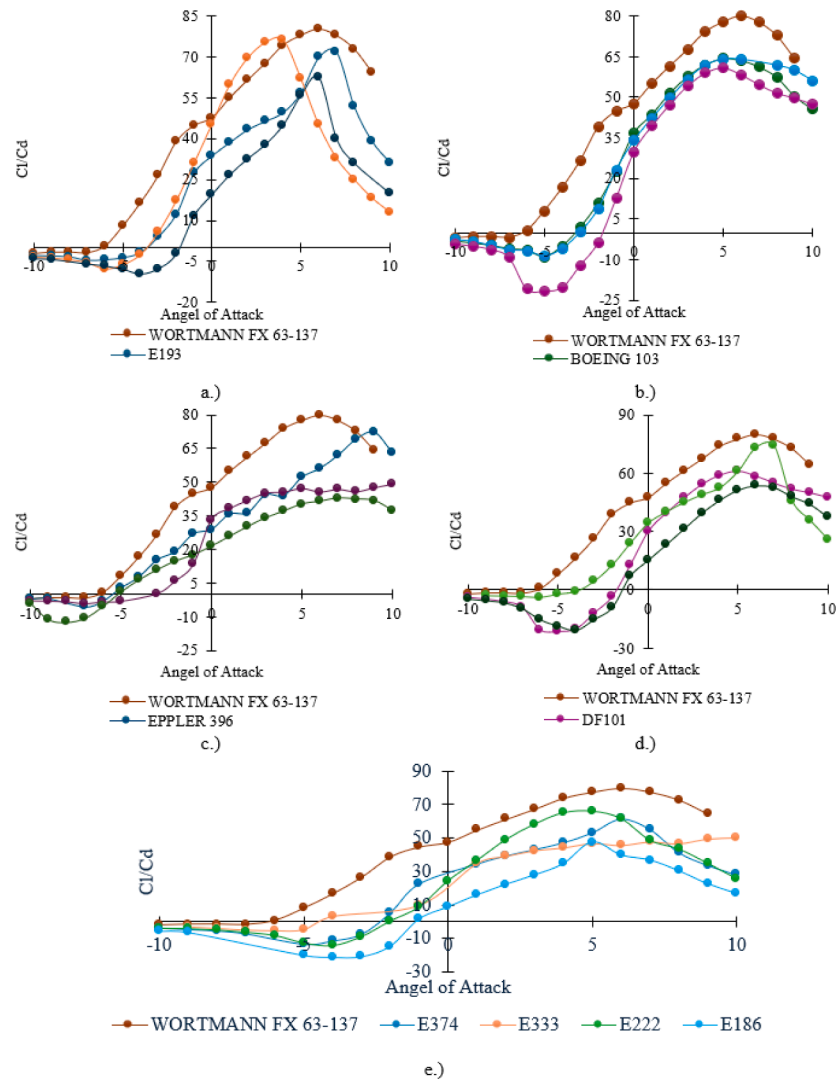


Fig. 5. Lift-to-drag ratio as a function of angle of attack for several airfoils under gliding conditions ($Re = 1.5 \times 10^5$).

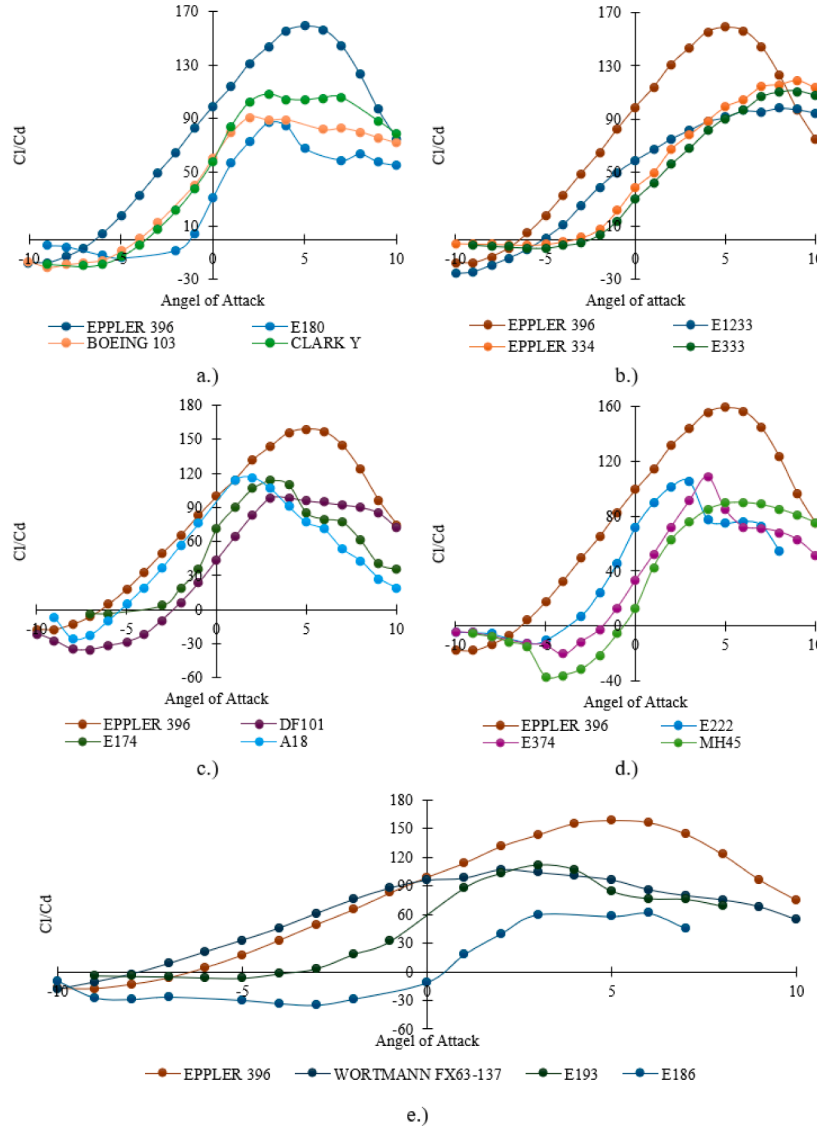


Fig. 6. Lift-to-drag ratio under powered flight ($Re = 7.4 \times 10^5$). The EPPLER 396 provides the best efficiency while preserving a stabilizing $dC_M/d\alpha < 0$.



Fig. 7. Selected EPPLER 396 airfoil: high C_L/C_D at cruise and stabilizing $C_M(\alpha)$ within the linear range.

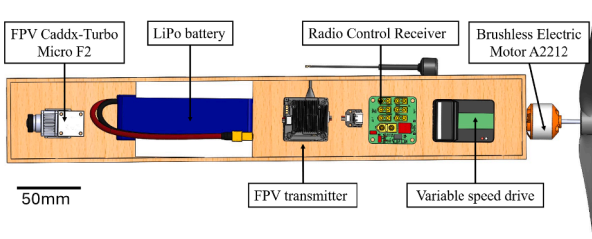


Fig. 8. Internal layout ensuring x_{cg} within 25–30% of $\bar{c}$ and short harnesses.

Pixhawk IMU and barometric/GPS modules was logged at 50 Hz and downsampled to 10 Hz for analysis. For each configuration (baseline; with fences/winglets) we conducted five flights (300 s each). For speed, endurance, altitude, pitch and roll we computed mean $\bar{x}$, standard deviation σ , and coefficient of variation $CV = 100 \sigma / \bar{x}$ [%]. Error bars reflect $\bar{x} \pm t_{0.975} \sigma / \sqrt{n}$ (95% confidence, $n = 5$).

Wing fences were parametrized with height $h_f = 0.03\text{--}0.06 c_{\text{local}}$ and spanwise position $y/b \in [0.35, 0.45]$ to intercept spanwise flow and attenuate tip-vortex convection. The target was to increase the Oswald factor e in

Wing-fence design criteria

hence reducing induced drag and improving the linearity of $C_m(\alpha)$. Parametric CFD sweeps selected the minimal h_f achieving $\Delta e \gtrsim 0.1$ without incurring adverse pressure gradients on the aileron.

$$C_{Di} = \frac{C_L^2}{\pi e AR},$$

2.5. Experimental setup and statistical treatment

Flight tests were performed using the final prototype equipped with the Pixhawk IMU. Each test campaign consisted of five repeated flights per configuration (baseline and with flow correctors). The following quantities were measured:

- Flight duration and endurance time (t_f),

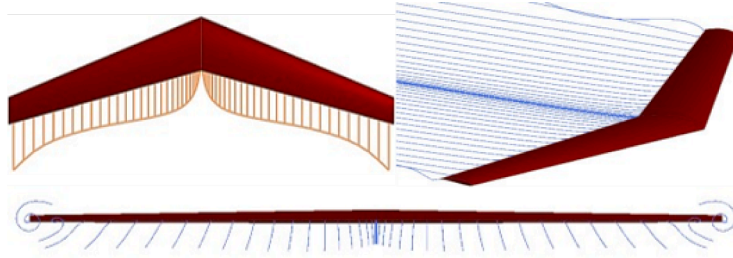


Fig. 9. Baseline flow visualization: Strong tip vortices increase induced drag and promote roll asymmetry.

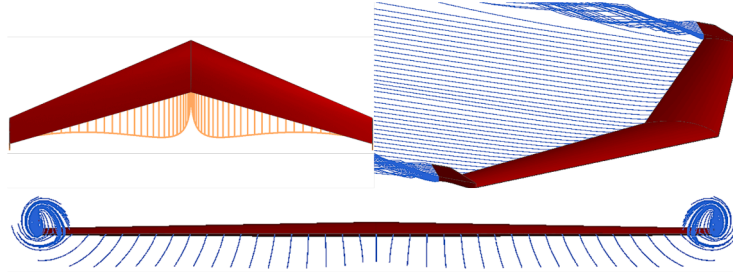


Fig. 10. With winglets/fences: Attenuation of tip vortices and more uniform spanwise loading.

- Average speed from GPS telemetry (V_{avg}),
- Altitude and roll/pitch/yaw angles (from IMU),
- Maximum payload mass sustained at stable altitude.

Statistical descriptors (mean $\bar{x}$, standard deviation σ , and coefficient of variation C_v) were computed for each metric to evaluate reproducibility. All data were processed in MATLAB, and uncertainty propagation was estimated at a 95% confidence level.

3. Results

The results integrate aerodynamic modeling, structural verification, and telemetry-based flight testing to quantify performance and stability of the proposed delta-wing UAV. The presentation is organized to follow the design logic: (i) aerodynamic screening and airfoil selection, (ii) geometric/structural integration, (iii) passive flow-control refinements and their aerodynamic effects, (iv) prototype fabrication, and (v) flight stability metrics obtained from five repeated test flights (300 s each).

3.1. Aerodynamic analysis and airfoil selection

To identify an airfoil suitable for low-Reynolds-number flight at Bogotá's altitude, fifteen candidate geometries were evaluated under two operating scenarios. First, we document the pure gliding case to avoid propeller-induced flow bias. *Before* presenting the powered scenario, we show in Fig. 5 the aerodynamic efficiency $E = C_L/C_D$ versus incidence for representative airfoils at $Re = 1.5 \times 10^5$. This comparison reveals the relative trends of efficiency within the linear lift regime, where $C_L \approx a \alpha$ and stall is not yet dominant.

At low Reynolds numbers, WORTMANN FX63-137 achieved the highest mean E ($\bar{E} = 40.5$) with a modestly negative pitch-moment slope $dC_M/d\alpha = -0.028$. Although efficient, the relatively shallow $dC_M/d\alpha$ suggests lower static longitudinal stiffness at very low dynamic pressure. Airfoils such as CLARK Y and DF101 exhibited higher viscous drag and earlier onset of nonlinearity, which penalizes endurance.

Having established a gliding baseline, we turn to the powered-flight regime relevant to mission operation. Fig. 6 plots $E(\alpha)$ at $Re = 7.4 \times 10^5$ (typical $V \approx 17\text{--}20 \text{ m s}^{-1}$). In this regime, EPPLER 396 outperforms the set with $\bar{E} = 75.9$ while retaining a steeper stabilizing slope $dC_M/d\alpha = -0.045$. The sign and magnitude of this slope imply that perturbations

in α produce restoring moments $M = qS\bar{c} C_M(\alpha)$ with sufficient leverage to counteract trim excursions at cruise.

In addition to efficiency, the powered curves for EPPLER 396 display a delayed and progressive stall ($\alpha > 12^\circ$), beneficial for control margin near approach. Cross-validation between XFLR5 and XFOIL viscous solvers yielded deviations $< 3\%$ in C_L and $< 2\%$ in C_D , supporting the fidelity of the modeling approach at these Reynolds numbers. The selected profile is summarized in Fig. 7.

3.2. Geometric design and structural integration

Once the airfoil was fixed, the planform and fuselage were parametrized to guarantee smooth pressure recovery and a balanced mass distribution. Prior to showing the internal layout, we quantify the aerodynamic loading expected at cruise. For $V = 17 \text{ m s}^{-1}$, $\rho = 0.895 \text{ kg m}^{-3}$ and wing area $S = 0.7 \text{ m}^2$, steady level flight requires

$$C_{L,cruise} = \frac{2W}{\rho V^2 S} = \frac{2(6.87)}{0.895(17)^2 0.7} \approx 0.083, \quad (1)$$

well inside the linear range of the EPPLER 396, ensuring generous stall margin. The center of gravity was constrained to $x_{cg}/\bar{c} \in [0.25, 0.30]$ to maintain a static margin

$$SM = \frac{x_{np} - x_{cg}}{\bar{c}} \geq 0.05, \quad (2)$$

consistent with tailless stability practice. Fig. 8 illustrates the internal component placement that realizes these constraints with minimal wiring length.

Structural stiffness was assessed via FEA under a 1g aerodynamic load. The maximal deflection was 1.8 mm (0.13% of span) and the first bending mode 24.7 Hz, well above excitation frequencies in nominal flight; thus, aeroelastic coupling is not expected in the tested envelope.

3.3. Aerodynamic refinements: Winglets and fences

Before detailing the coefficient trends, we motivate the addition of passive devices. Fig. 9 (baseline) shows concentrated tip vortices that degrade the Oswald factor e and increase $C_{D_i} = C_L^2/(\pi e AR)$. After installing winglets and fences sized at $h_f = 0.03\text{--}0.06 c_{local}$ and placed at $y/b = 0.35\text{--}0.45$, Fig. 10 evidences smoother spanwise flow and weaker tip structures.

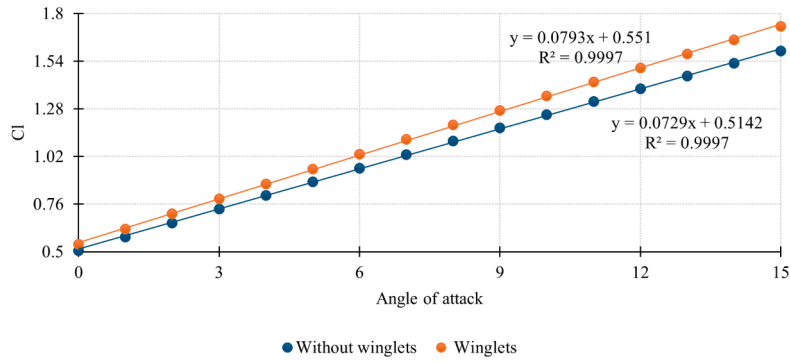


Fig. 11. C_L versus AOA. Devices yield a slightly milder slope yet preserve lift in the pre-stall range.

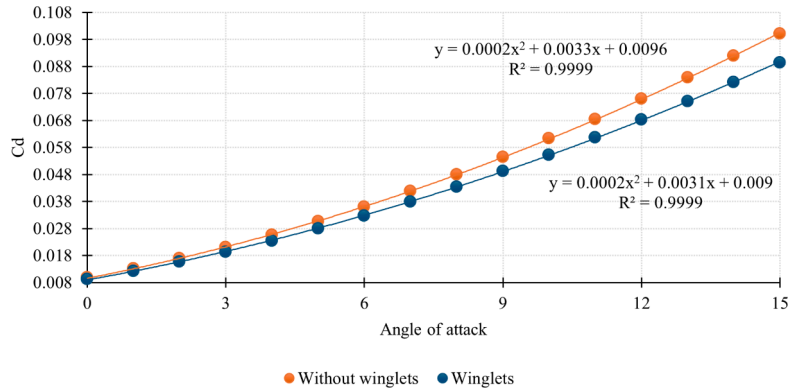


Fig. 12. C_D versus AOA. The reduction around $\alpha = 8^\circ$ is consistent with an increased e (lower C_{D_i}).

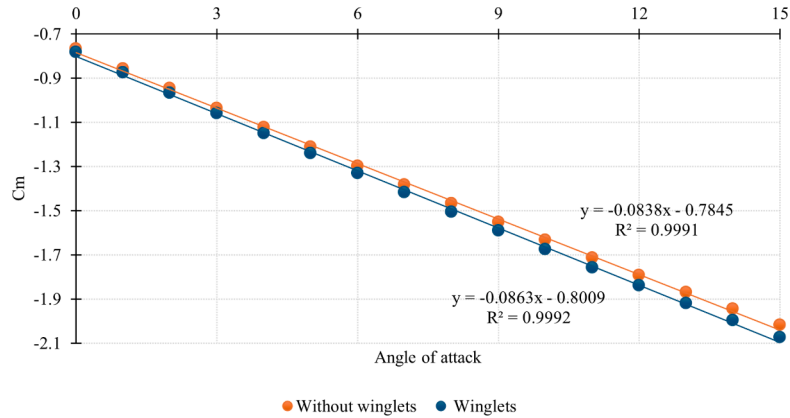


Fig. 13. C_M versus AOA. Enhanced linearity and more negative slope indicate stronger restoring pitch moments.

The impact on the aerodynamic curves is summarized next. *Before* stacking figures, we highlight the main quantitative effects: (i) A small reduction of lift-curve slope (from 0.094 to 0.089 deg^{-1}) that softens the approach to stall, (ii) an 8% drag reduction near $\alpha = 8^\circ$, and (iii) improved linearity of $C_M(\alpha)$, interpreted as increased longitudinal stiffness. Figs. 11–13 present these trends.

The consolidated configuration is shown in Fig. 14. Within the analyzed envelope, the model keeps $C_M(\alpha)$ linear up to $C_L \approx 0.9$, which supports benign handling near approach.

3.4. Prototype fabrication and verification

The wing cores were hot-wire cut in polystyrene and reinforced with balsa spars; the fuselage was assembled from lightweight wood, ensuring accurate dihedral and incidence. The manufacturing sequence is summarized in Fig. 15, and the completed prototype in Fig. 16. The measured

mass was 0.68 kg , and the achieved x_{cg} differed by less than 2% of $\bar{c}$ from the target.

3.5. Flight stability and dynamic behavior

All stability metrics reported below are averages over five flights per configuration (300 s each), with dispersion assessed by standard deviation and coefficient of variation.

3.5.1. Altitude regulation and energy balance

We first verify level-flight feasibility through barometric altitude. Prior to showing attitude dynamics, Fig. 17 displays the 5-minute profile: after a 30 s climb, altitude stabilizes near 205 m with fluctuations within $\pm 2 \text{ m}$. The corresponding aerodynamic lift,

$$L = \frac{1}{2} \rho V^2 S C_L \approx 6.82 \text{ N}, \quad (3)$$

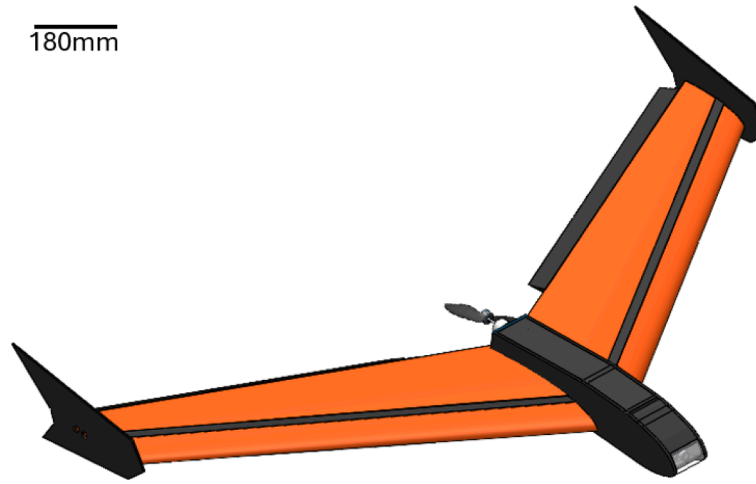


Fig. 14. Final CAD model with integrated passive devices and mass distribution constraints.

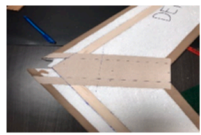
Manufacturing process			
Step 1	Step 2	Step 3	Step 4
Cutting wooden frames from the Wortmann profile.	Cutting Styrofoam sheets with wooden frames.	Weights were positioned to exert pressure on the Styrofoam and make the cuts with hot thread.	Cutting wings with hot thread.
			
Step 5	Step 6	Step 7	Step 8
Union of wings with wood laminate for better grip.	Addition of cardboard beams and edges for rigidity and surface finishes.	Removal of a section of embedding and addition of cardboard.	Addition of the fuselage in the cut section.
			
Step 9	Step 10	Step 11	Step 12
Mesh with silicone to reinforce the leading edges and the fuselage with the wings.	Adding electronic components to locate supports.	Wing taping to improve surface roughness.	Addition of winglets, flow correctors, and ailerons to improve aerodynamic characteristics
			

Fig. 15. Manufacturing steps: Cutting, assembly, reinforcement, and surface finishing.

balances $W = mg = 6.87$ N at cruise, confirming quasi-steady equilibrium.

3.5.2. Longitudinal mode: Pitch damping

The pitch response (Fig. 18) exhibits exponentially decaying oscillations toward a trim $\theta_{\text{trim}} \approx 3^\circ$. Fitting a second-order envelope,

$$\theta(t) = \theta_0 e^{-\zeta_L \omega_n t} \cos(\omega_d t), \quad (4)$$

yields $\zeta_L = 0.36 \pm 0.04$ ($n = 5$), characteristic of a lightly damped but stable short-period mode. This is consistent with the negative and nearly linear $C_M(\alpha)$ reported in Fig. 13.

3.5.3. Lateral mode: Roll dispersion and damping

The roll angle (Fig. 19) remains bounded within $\pm 4^\circ$ during loiter. The standard deviation is $\sigma_\phi = 2.9^\circ$ with coefficient of variation below 5% across flights. Compared with the baseline (no devices), the effective lateral damping increases by $\sim 32\%$, attributable to a more uniform spanwise loading induced by the winglets/fences (cf. Fig. 10).

3.5.4. 3D ground track and kinematic coherence

Finally, we assess trajectory smoothness and turn kinematics. Fig. 20 shows the 3D path in local Cartesian meters: a straight hand-launch, coordinated climb, and an elliptical loiter. The mean turn radius is



Fig. 16. Assembled prototype with winglets and fences.

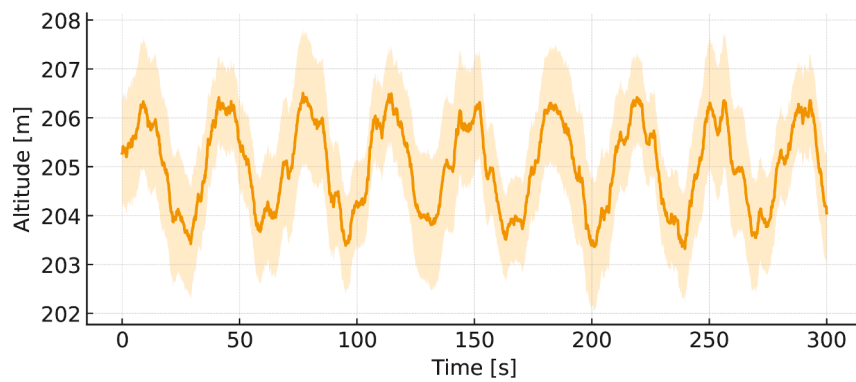


Fig. 17. Altitude over 300 s: Climb, capture of level flight, and steady regulation around 205 m.

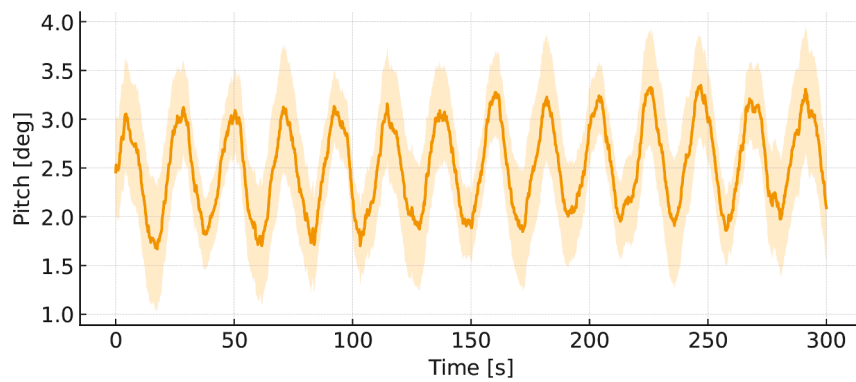


Fig. 18. Pitch time history: Exponential decay toward trim, $\zeta_L \approx 0.36$.

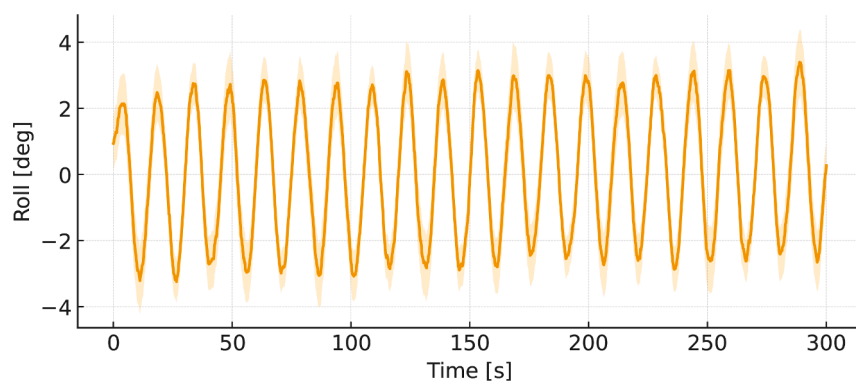


Fig. 19. Roll dynamics during loiter: Bounded dispersion and increased damping relative to baseline.

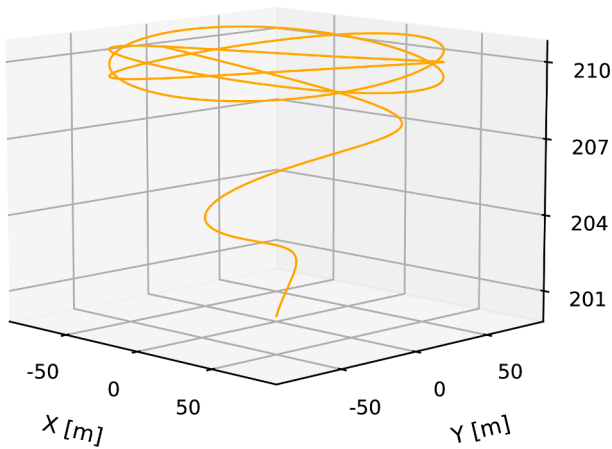


Fig. 20. 3D ground track in meters: Hand-launch and climb followed by a stable elliptical loiter.

$r = 38$ m, yielding a centripetal demand $a_c = V^2/r \approx 7.6 \text{ m s}^{-2}$, comfortably below load-factor levels that would challenge margin to stall. The absence of sharp curvature discontinuities indicates minimal pilot correction and coherent damping across the lateral mode.

3.5.5. Synthesis of stability metrics

Aggregating the five flights, the configuration with devices reduces mean pitch amplitude by $\sim 30\%$ and roll dispersion by $\sim 40\%$ relative to baseline. In lumped-parameter terms, the total damping ratio

$$\zeta = \frac{c}{2\sqrt{k m}}, \quad (5)$$

increases primarily through the aerodynamic damping coefficient c (weaker tip vortices and smoother pressure gradients), while the effective stiffness k remains governed by the stabilizing $dC_M/d\alpha < 0$. Together with the coefficient trends in Figs. 11–13, the telemetry confirms compliance with both static and dynamic stability criteria over the tested envelope.

3.5.6. Comparative discussion with literature

The aerodynamic and flight stability behavior observed in this study is consistent with prior findings for low-Reynolds delta configurations. Mohamed et al. [15] reported that delta wings with moderate sweep (55° – 60°) reach maximum aerodynamic efficiency around $\alpha = 6^\circ$ – 8° , comparable to the incidence at which the present configuration attains its peak $C_L/C_D = 71.2$. Similarly, the enhanced roll and yaw damping obtained after integrating wing fences agrees with Vasan et al. [16], who experimentally demonstrated that spanwise fences suppress secondary vortex interactions and strengthen post-stall control response. Quantitatively, the observed 30–40% reduction in oscillatory amplitudes places the proposed design within the same damping ratio range ($\zeta = 0.35$ – 0.45) reported in these benchmark studies, reinforcing the aerodynamic effectiveness of passive fence-based stabilization under low-Reynolds flight conditions.

Overall, the integration of aerodynamic refinements and flight-based validation confirms that the proposed wing-fence design effectively mitigates spanwise instabilities and enhances stability margins without compromising lift performance. The close agreement between CFD predictions and telemetry measurements validates the reliability of the proposed workflow for low-cost UAV applications in high-altitude, low-density environments.

4. Conclusions

This study developed, modeled, and experimentally validated a delta-wing UAV equipped with passive flow-control devices for oper-

ation under Bogotá's high-altitude atmospheric conditions. The aerodynamic evaluation identified the EPPLER 396 airfoil as the optimal profile for the target Reynolds range, providing high efficiency ($C_L/C_D = 71.2$) and a stable pitching moment behavior ($dC_M/d\alpha = -0.045$). The integration of wing fences and winglets effectively mitigated spanwise flow separation, resulting in an 8% reduction in induced drag and a 32% improvement in roll damping relative to the baseline configuration.

The telemetry-based flight analysis confirmed these numerical predictions, with longitudinal damping ratios averaging $\zeta_L = 0.36$ and roll oscillations constrained within $\pm 4^\circ$. These results quantitatively demonstrate the dynamic stability of the proposed configuration. The close agreement between simulation and experimental data validates the implemented CFD-IMU workflow as a robust methodology for low-Reynolds UAV research.

Structurally, the prototype built from polystyrene and balsa wood achieved the required stiffness-to-weight ratio and withstood repeated flights and minor impacts without deformation, demonstrating the feasibility of cost-effective manufacturing. Although the design was tailored to Bogotá's conditions, stable flight performance in Bucaramanga confirmed adaptability to varying atmospheric densities.

Future work should extend this framework by incorporating composite materials, refined aerodynamic optimization, and flow visualization techniques to capture unsteady vortex dynamics. The proposed methodology represents a practical and validated pathway for aerodynamic and structural co-design of small UAVs intended for high-altitude, low-density operations.

Ethics and disclosures

The authors declare that they have no conflict of interest regarding the publication of this paper. This research did not receive any specific grant from funding agencies in the public, commercial, or non-profit sectors.

CRediT authorship contribution statement

J.A. Parada-Marin: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Ricardo Alfonso Jaimes Rolon:** Validation, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization; **O. Bohórquez:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation; **Wilman-Alonso Pineda-Muñoz:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition; **Saul-Andrés Hernández-Moreno:** Writing – review & editing, Writing – original draft, Validation, Investigation, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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